
STUDY GUIDE

Renal Physiology

Filtration · Reabsorption · Secretion · Water Balance ·
Acid-Base

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PHS 220 — Physiology

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§1 ANATOMY OF THE KIDNEY

The kidneys are paired, bean-shaped organs ($\sim 11 \times 6 \times 3$ cm; ~ 150 g each) lying retroperitoneally in the posterior abdominal wall at the level of T12-L3. The right kidney sits slightly lower than the left (due to the liver above it). Despite representing only 0.5% of body mass, the kidneys receive approximately 20–25% of cardiac output (~ 1.2 L/min) — the highest perfusion rate per gram of any organ.

Feature	Description
Coverings	True capsule (fibrous); Perirenal fat; Renal (Gerota's) fascia; Pararenal fat. Each layer separable in surgical anatomy.
Hilum	Medial concavity; entry/exit for renal vein (anterior), renal artery (middle), ureter (posterior) — "VAU" from anterior to posterior
Cortex	Outer reddish-brown granular zone (~ 1 cm thick); contains glomeruli, proximal and distal tubules, Malpighian (renal) corpuscles; interdigitates as "cortical columns" (Columns of Bertin) between pyramids
Medulla	Inner pale zone; 8–18 renal pyramids per kidney; striations due to parallel loops of Henle and collecting ducts; contains thick and thin limbs of Henle and collecting ducts
Renal Papillae	Apex of each pyramid; 15–20 papillary ducts (of Bellini) open at each papilla → drain into minor calyx. Site where urine enters the collecting system.
Calyces	7–14 minor calyces (each cups one papilla) → 2–3 major calyces → renal pelvis → ureter → bladder
Arterial supply	Renal artery (from abdominal aorta, L1–L2) → segmental arteries (end arteries: no collaterals — infarction = wedge-shaped) → interlobar → arcuate → interlobular → afferent arterioles → glomerular capillaries → efferent arterioles → peritubular capillaries OR vasa recta (juxtamedullary)

§2 THE NEPHRON — FUNCTIONAL UNIT

The nephron is the functional unit of the kidney. Each kidney contains approximately 1–1.5 million nephrons, none of which regenerate after birth. Loss of nephrons — through disease, age, or surgery — results in compensatory hypertrophy of the remaining nephrons, which can maintain adequate function until significant loss has occurred.

2.1 Types of Nephron

Type	Glomerulus Location	% of Nephrons	Loop of Henle	Function
Cortical nephrons	Outer and mid cortex	$\sim 85\%$	Short loop; turns in outer medulla or at cortico-medullary junction	Filtration and tubular function; less important for urine concentration

Type	Glomerulus Location	% of Nephrons	Loop of Henle	Function
Juxtamedullary nephrons	Inner cortex (near cortico-medullary junction)	~15%	Long loop; descends deep into inner medulla; associated with vasa recta	Countercurrent multiplication and exchange; essential for production of concentrated urine

2.2 Renal Corpuscle (Glomerulus)

The renal corpuscle consists of the glomerular tuft (a knot of specialised fenestrated capillaries) surrounded by Bowman's capsule. Filtration occurs from the glomerular capillary lumen across the filtration barrier into Bowman's space (urinary space), producing the primary filtrate (ultrafiltrate of plasma).

Component	Structure	Function/Contribution to Filtration
Fenestrated endothelium	Large pores (70–100 nm); no diaphragm; negatively charged glycocalyx	Permits free passage of water, solutes, small-medium proteins; restricts blood cells; charge barrier partially excludes anionic proteins
Glomerular Basement Membrane (GBM)	Fused endothelial and podocyte basement membranes; collagen IV, laminin, heparan sulphate proteoglycans; thickness ~300 nm	Primary size and charge barrier; negative charge (HSPGs) repels negatively charged albumin; GBM thickening in diabetes → initially proteinuria
Podocytes	Visceral epithelial cells; "foot processes" (pedicles) interdigitate; filtration slit diaphragm (FSD) between adjacent foot processes contains nephrin and podocin	FSD is the final barrier; diameter ~25–60 nm; loss/fusion of foot processes (minimal change disease) → massive proteinuria; nephrin mutations → congenital nephrotic syndrome
Mesangial Cells	Between glomerular capillaries; contractile; phagocytic	Structural support; regulate capillary surface area (Ang II → contraction → ↓ filtration surface → ↓ GFR); phagocytose macromolecules; produce cytokines
Parietal Epithelium	Flat epithelial cells lining Bowman's capsule; transition to tubular epithelium at neck	Forms outer wall of filtration space; connects to PCT at urinary pole

2.3 Tubular Segments

Segment	Location	Epithelium	Primary Functions
Proximal Convoluted Tubule (PCT)	Cortex; coils around glomerulus	Cuboidal with brush border (microvilli ↑ SA ×20); leaky tight junctions; rich mitochondria	Reabsorbs 65–70% of filtered Na^+ , water, HCO_3^- , glucose, AAs, K^+ , phosphate; secretes H^+ , organic acids/bases, drugs

Segment	Location	Epithelium	Primary Functions
Proximal Straight Tubule (PST / pars recta)	Descends into medullary ray	Similar to PCT but fewer mitochondria	Continues PCT functions; more secretion of organic anions (OAT1/3)
Thin Descending Limb of Henle (TDL)	Descends into inner medulla	Flat squamous; very few mitochondria; high AQP1 expression	Highly permeable to water; relatively impermeable to Na^+ → water leaves as it descends into hypertonic medulla → tubular fluid becomes concentrated
Thin Ascending Limb (TAL — inner)	Inner medulla; bends to become thick ascending	Flat squamous; no AQP; ion channels	Permeable to NaCl ; impermeable to water → NaCl moves out passively (dilutes tubular fluid, adds to medullary gradient)
Thick Ascending Limb (TAL — outer)	Outer medulla → cortex; Macula densa at end	Cuboidal; abundant mitochondria; NO AQP; NKCC2 transporter (target of loop diuretics)	NKCC2: active transport of Na^+ , K^+ , 2Cl^- out of lumen; impermeable to water → dilution of tubular fluid ("diluting segment"); generates medullary interstitial gradient
Macula Densa	At end of TAL / transition to DCT; adjacent to glomerulus	Dense (short, closely packed cells)	Senses luminal $[\text{NaCl}]$ → tubuloglomerular feedback (TGF) → signals JGA to adjust GFR; controls renin release
Distal Convoluted Tubule (DCT)	Cortex	Cuboidal; NCC (Na-Cl co-transporter — thiazide target); tight junctions	Reabsorbs ~5–10% Na^+ (via NCC); Ca^{2+} reabsorption (PTH-stimulated, TRPV5); regulated Na^+ reabsorption
Connecting Tubule (CNT)	Cortex; between DCT and CCD	Mixture of DCT and CD cells	Transitional; ENaC and ROMK present; aldosterone sensitive
Cortical Collecting Duct (CCD)	Cortex; merges to form outer medullary CD	Principal cells (Na reabsorption/ K secretion via ENaC/ROMK — aldosterone target); Intercalated cells ($\text{H}^+/\text{HCO}_3^-$ transport — acid-base)	Fine-tuning of Na^+/K^+ balance (aldosterone); acid-base regulation; ADH-independent water handling here
Inner Medullary Collecting Duct (IMCD)	Inner medulla; opens at papilla	Principal cells (AQP2 — ADH regulated); urea transporters (UT-A1/3 — ADH regulated)	Final concentration of urine (ADH-dependent water reabsorption; urea recycling); urinary acidification

§3 GLOMERULAR FILTRATION

Glomerular filtration is the process by which plasma is forced from the glomerular capillaries across the filtration barrier into Bowman's space, producing an ultrafiltrate of plasma. This ultrafiltrate lacks cells and large proteins but otherwise closely resembles plasma in composition.

3.1 Filtration Barrier Selectivity

Substance	MW (kDa)	Filterability	Reason
Na ⁺ , K ⁺ , Cl ⁻ , glucose, urea, creatinine	<0.5 kDa	~1.0 (freely filtered)	Small, uncharged; unrestricted passage
Myoglobin	17 kDa	~0.75	Small protein; not charge-restricted
Albumin	69 kDa	~0.001 (minimally filtered)	Size + strong negative charge (anionic) → repelled by GBM (heparan sulphate) + FSD
IgG	150 kDa	<0.001	Too large for filtration barrier
IgM	900 kDa	~0 (not filtered)	Pentameric — far too large

3.2 Starling Forces Governing Filtration

The driving force for glomerular filtration is the **Net Filtration Pressure (NFP)**, governed by the modified Starling equation, similar to that in peripheral capillaries but with unique quantitative values:

Force	Sym bol	Normal Value (mmHg)	Direction	Effect
Glomerular capillary hydrostatic pressure	PGC	~60 mmHg (high — maintained by efferent arteriolar resistance)	Outward (favours filtration)	Primary driving force for filtration; reduced by: ↓BP, afferent vasoconstriction, NSAIDs (block PGE2 vasodilation)
Bowman's space hydrostatic pressure	PBS	~15 mmHg	Inward (opposes filtration)	Increased by: ureteral obstruction → hydronephrosis → ↑PBS → ↓GFR
Glomerular capillary oncotic pressure	πGC	~25-30 mmHg (rises from 28→36 along capillary as protein concentrates)	Inward (opposes filtration)	Increased by: hypoproteinaemia → ↓πGC → ↑GFR; dehydration → ↑πGC → ↓GFR

Force	Sym bol	Normal Value (mmHg)	Direction	Effect
Bowman's space oncotic pressure	π_{BS}	~0-2 mmHg (minimal protein in normal filtrate)	Outward (favours filtration)	Rises dramatically in glomerulonephritis → ↓ filtration driving force

NET FILTRATION PRESSURE & GFR

$$NFP = (PGC + \pi_{BS}) - (PBS + \pi_{GC})$$

$$= (60 + 2) - (15 + 28) = 62 - 43 = \sim 19 \text{ mmHg net outward (early in glomerular capillary).}$$

GFR = $K_f \times NFP$, where K_f = ultrafiltration coefficient (product of hydraulic conductance and filtration surface area).

Normal GFR: ~125 mL/min (~180 L/day filtered) in a 70-kg adult man; ~110 mL/min in adult woman. Of this, only 1-2 L/day exits as urine (>99% reabsorbed).

§4 FACTORS REGULATING GFR

Regulatory Mechanism	Description	Effect on GFR
Renal Autoregulation (Myogenic)	Afferent arteriole smooth muscle contracts when wall tension increases (↑ stretch from ↑ BP); keeps GFR constant despite MAP 80-180 mmHg	Maintains GFR constant; fails below MAP ~80 mmHg → prerenal AKI
Tubuloglomerular Feedback (TGF)	Macula densa senses ↑ NaCl delivery → releases adenosine → afferent arteriole vasoconstriction → ↓ GFR → ↓ NaCl delivery (self-limiting). This is the other component of autoregulation.	↑ NaCl delivery → ↓ GFR (A1 receptor; adenosine/ATP); ↓ NaCl delivery → ↑ GFR
Sympathetic NS (α_1 receptors)	Severe stress → renal vasoconstriction of afferent (and to a lesser extent efferent) arterioles → ↓ renal blood flow and GFR	↓ GFR; diverts blood from kidney in emergency
Angiotensin II	Preferentially constricts EFFERENT arteriole → ↑ PGC → initially maintains GFR despite ↓ RBF. ACE inhibitors/ARBs dilate efferent → ↓ PGC → ↓ GFR (but also ↓ proteinuria — anti-fibrotic).	Maintains GFR in low-flow states; ACEi block → ↓ GFR especially with RAS/bilateral RAS
Prostaglandins (PGE ₂ , PGI ₂)	Afferent arteriole vasodilators; counteract angiotensin II vasoconstriction; important in maintaining RBF/GFR in low-flow states (heart failure, cirrhosis, volume depletion)	↑ RBF and GFR (protective). NSAIDs block PGs → efferent dilation lost → ↑ Risk of AKI in low-flow states
ANP/BNP	Dilates afferent; constricts efferent → ↑ PGC; also dilates mesangial cells (↑ K_f)	↑ GFR; ↑ sodium excretion (natriuresis)

Regulatory Mechanism	Description	Effect on GFR
Dopamine (low dose)	D1 receptors on renal vasculature → vasodilation	↑ RBF and GFR at low doses (renal dose dopamine)

§5 TUBULAR REABSORPTION

Of the 180 L of filtrate produced daily, only 1-2 L exits as urine. The remaining 178+ L is reabsorbed across the tubular epithelium back into peritubular capillaries. Reabsorption is a selective, active and passive process, regulated by hormones and the body's current needs.

Substance	Filtered/day	Excreted/day	% Reabsorbed	Primary Site(s)
Water	180 L	1.5 L	>99%	PCT (67%, AQP1); TAL (nil — impermeable); CD (variable — ADH-dependent, AQP2)
Na ⁺	25,200 mmol	~100 mmol	>99%	PCT (67%); TAL/NKCC2 (25%); DCT/NCC (5%); CD/ENaC (2-3% — fine-tuned by aldosterone)
Cl ⁻	18,000 mmol	~100 mmol	>99%	Follows Na ⁺ in PCT (paracellular); NKCC2 in TAL; NCC in DCT
K ⁺	720 mmol	~70 mmol	~90% net (but ~90% reabsorbed PCT/TAL, then 10-60% secreted in CD)	PCT (67%, K-H ATPase + paracellular); TAL (NKCC2 + ROMK recycling); CD (secretion via ROMK — aldosterone-stimulated)
HCO ₃ ⁻	4,500 mmol	~2 mmol	>99%	PCT (80-90%); TAL (5%); CD (fine regulation, intercalated cells)
Glucose	~180 g (all filtered; plasma ~5 mmol/L × 180L)	~0 g	100%	PCT entirely via SGLT2 (90%) and SGLT1 (10%); Tm ~375 mg/min → splay → glycosuria at plasma >~10-11 mmol/L
Urea	~54 g	~28 g	~40-60%	PCT (50%, passive); IMCD (recycled via UT-A1/3 — ADH-stimulated → contributes to medullary gradient); excreted: the rest
Phosphate	~1800 mg	~600 mg	~85%	PCT (NaPi-2a co-transporter; inhibited by PTH and FGF-23)
Calcium	~10,000 mg	~200 mg	~98%	PCT (70%, passive paracellular); TAL (20%, paracellular, driven by positive lumen); DCT (8-10%, active transcellular TRPV5 — stimulated by PTH and calcitriol)

GLUCOSE REABSORPTION & SGLT2 INHIBITORS

Glucose is reabsorbed entirely in the PCT (normally). SGLT2 handles 90% of glucose reabsorption (Na-glucose co-transport, 1Na:1Glucose) in the early PCT. SGLT1 handles the remaining 10% in the late PCT (2Na:1Glucose).

Transport Maximum (T_m): ~375 mg/min (males); ~300 mg/min (females). Once plasma glucose exceeds ~11 mmol/L (renal threshold), glucose appears in urine (glycosuria). In DM, persistent hyperglycaemia → osmotic diuresis → polyuria, polydipsia.

SGLT2 Inhibitors (Gliflozins — empagliflozin, dapagliflozin): Block SGLT2 → glycosuria even at normal plasma glucose → ↓ HbA1c; also cause osmotic diuresis → ↓ blood pressure; reduce intra-glomerular pressure; nephroprotective and cardioprotective (EMPA-REG, CREDENCE trials).

§6 TUBULAR SECRETION

Tubular secretion is the process by which substances are transported from the peritubular capillaries into the tubular lumen, augmenting their excretion beyond what filtration alone could achieve. This is particularly important for drugs, organic acids and bases, and endogenous metabolites that are extensively protein-bound (and therefore poorly filtered).

Substance Secreted	Site	Transporter	Clinical Application
K ⁺	Cortical collecting duct	ROMK channels (apical); Na/K-ATPase (basolateral)	Aldosterone stimulates K ⁺ secretion; hyperkalaemia from ACEi/spironolactone/K-sparing diuretics
H ⁺	PCT (Na/H exchanger, NHE3); I ntercalated cells (CD)	NHE3; H-ATPase; H/K-ATPase (type B intercalated)	Essential for acid-base regulation; produces titratable acid and ammonium
Creatinine	PCT (small amount ~10%)	Organic cation transporters	Creatinine clearance overestimates GFR by ~10–20% (secretion adds to filtered creatinine)
PAH (para-aminohippuric acid)	PCT (nearly completely)	OAT1/OAT3	PAH clearance used to measure effective renal plasma flow (ERPF) — ~90–95% extracted in one pass at low concentrations
Organic acids (uric acid, bile salts, penicillin, furosemide, probenecid target)	PCT	OAT1, OAT3 (basolateral); MRP2, ABCG2 (apical)	Probenecid blocks urate secretion AND tubular secretion of penicillin (historical clinical use)
Organic bases (morphine, dopamine, amiloride, cimetidine, quinine)	PCT	OCT2 (basolateral); MATE1/2 (apical)	Metformin secretion by OCT2 + MATE; reduced by cimetidine competition

§7 COUNTERCURRENT MULTIPLICATION & URINE CONCENTRATION

The ability to produce urine more concentrated than plasma (up to 1,200 mOsmol/kg in humans) depends on the generation of a hyperosmotic medullary interstitium by the **countercurrent multiplier** (loop of Henle) and maintenance of the gradient by the **countercurrent exchanger** (vasa recta).

Component	Mechanism	Contribution to Medullary Gradient
Thick Ascending Limb (TAL)	Actively pumps NaCl out of tubular lumen into interstitium (NKCC2 + Na/K-ATPase); WATER CANNOT FOLLOW (impermeable to water) → lumen dilutes, interstitium concentrates	Primary source of hyperosmolality in outer medulla (~600–800 mOsmol/kg); "single effect" driving countercurrent multiplication
Thin Descending Limb (TDL)	Permeable to water (AQP1); relatively impermeable to NaCl; water moves OUT as tubular fluid enters hypertonic medulla → tubular fluid becomes concentrated before turning	Concentrates tubular fluid (delivers more NaCl to thick TAL); amplifies single effect by countercurrent arrangement
Urea Recycling (ADH-stimulated)	ADH → UT-A1/3 in inner medullary CD → urea secreted into inner medullary interstitium; UT-A2 in thin descending limb → urea re-enters tubule → recycled	40–50% of inner medullary gradient is contributed by urea; explains why protein-deficient diet impairs maximal urine concentration
Vasa Recta (Countercurrent Exchange)	Descending vasa recta permeable to solute/water; ascending vasa recta returns solute to deep medulla; parallel flow (descending = blood going deep; ascending = returning) allows exchange without dissipating gradient	Maintains gradient established by loop; does NOT generate gradient; if absent → gradient washed out. Reduces solute removal from medulla.

7.1 ADH-Dependent Concentration in Collecting Duct

ADH Level	AQP2 Expression	IMCD Water Permeability	Urine Output	Urine Osmolality
Maximum (concentrated urine)	High (AQP2 inserted into apical membrane)	Very high	~0.5 L/day (minimum)	~1200 mOsmol/kg (max concentration)
Absent (maximally dilute urine)	Low (AQP2 retrieved into cytosolic vesicles)	Very low	~20 L/day (maximum)	~50 mOsmol/kg (max dilution)

§8 REGULATION OF WATER BALANCE (ADH / VASOPRESSIN)

Aspect	Details
Synthesis	Nonapeptide (9 amino acids); synthesised in supraoptic (SON) and paraventricular (PVN) nuclei of hypothalamus → transported down axons → stored in posterior pituitary

Aspect	Details
Primary stimulus	Plasma hyperosmolality — osmoreceptors in anterior hypothalamus (also circumventricular organs — outside BBB) detect $\uparrow$ effective osmolality $\rightarrow$ ADH release. Threshold: ~ 280 mOsmol/kg. 1 mOsmol/kg rise $\rightarrow$ 1 pg/mL rise in plasma ADH.
Secondary stimulus	Hypovolaemia (>8 -10% volume loss) $\rightarrow$ low-pressure baroreceptors (atria/pulmonary) + high-pressure baroreceptors (carotid) $\rightarrow$ overrides osmotic set point $\rightarrow$ ADH released even if hypo-osmolar (e.g., SIADH-like effect in severe haemorrhage)
ADH Receptor — V2	Gs-coupled GPCR in principal cells of CD $\rightarrow$ $\uparrow$ cAMP $\rightarrow$ PKA $\rightarrow$ AQP2 vesicles translocate to apical membrane $\rightarrow$ water permeability $\uparrow$ $\rightarrow$ water reabsorbed
ADH Receptor — V1a	Gq-coupled on vascular smooth muscle $\rightarrow$ IP3/Ca ²⁺ $\rightarrow$ vasoconstriction (vasopressor effect) — important in haemorrhagic shock

Disorder	Mechanism	Plasma Osmolality	Urine Osmolality	Clinical Features
Diabetes Insipidus — Central (CDI)	Insufficient ADH secretion (trauma, surgery, tumour, infiltration of SON/PVN)	HIGH (>295 mOsmol/kg)	LOW (<300 ; dilute)	Polyuria (up to 20L/day); polydipsia; responds to desmopressin (DDAVP)
Diabetes Insipidus — Nephrogenic (NDI)	Renal resistance to ADH (V2R mutation; aquaporin-2 mutation; lithium toxicity; hypercalcaemia; hypokalaemia)	HIGH	LOW (<300)	Same as CDI; does NOT respond to desmopressin
SIADH (Syndrome of Inappropriate ADH)	Excess ADH secretion (CNS disease, malignancy — SCLC, drugs — SSRIs, carbamazepine)	LOW (<280 mOsmol/kg — hyponatraemia)	HIGH (>100 ; inappropriately concentrated)	Hyponatraemia; nausea; confusion; seizures. Treat: fluid restriction; salt/loop diuretics; V2R antagonists (tolvaptan)

§9 REGULATION OF SODIUM & EXTRACELLULAR VOLUME

9.1 The RAAS — Renin-Angiotensin-Aldosterone System

- $\downarrow$ Renal perfusion pressure $\rightarrow$ juxtaglomerular cells (modified smooth muscle in afferent arteriole) release **Renin** (also stimulated by $\uparrow$ SNS β_1 ; $\downarrow$ NaCl at macula densa).
- Renin cleaves **Angiotensinogen** (synthesised by liver) $\rightarrow$ **Angiotensin I** (decapeptide, inactive).
- Angiotensin-Converting Enzyme (ACE) in pulmonary capillary endothelium (mainly) cleaves 2 amino acids from Ang I $\rightarrow$ **Angiotensin II** (octapeptide, potent).
- Angiotensin II** acts via AT1 receptors: (a) Adrenal zona glomerulosa $\rightarrow$ Aldosterone release; (b) Efferent arteriole vasoconstriction $\rightarrow$ maintains GFR; (c) PCT NHE3 stimulation $\rightarrow$ $\uparrow$ Na⁺ reabsorption; (d) Thirst; (e) ADH secretion; (f) Sympathetic activation.
- Aldosterone** $\rightarrow$ mineralocorticoid receptor in principal cells of CD $\rightarrow$ $\uparrow$ ENaC (Na⁺ reabsorption) + $\uparrow$ ROMK (K⁺ secretion) $\rightarrow$ Na⁺ retained; K⁺ lost; water follows Na⁺ $\rightarrow$ $\uparrow$ ECF volume.

- Retained Na^+ and water $\rightarrow$ $\uparrow$ blood volume $\rightarrow$ $\uparrow$ BP $\rightarrow$ $\downarrow$ renin release (negative feedback).

RAAS Blocker	Mechanism	Effect	Clinical Use
ACE inhibitors (e.g., ramipril, lisinopril)	Block ACE $\rightarrow$ $\downarrow$ Ang II	$\downarrow$ BP; $\downarrow$ aldosterone; $\uparrow$ bradykinin ($\rightarrow$ cough, angio-oedema); $\downarrow$ proteinuria; $\downarrow$ efferent tone $\rightarrow$ $\downarrow$ GFR (caution in bilateral RAS)	Hypertension; HFrEF; diabetic nephropathy; post-MI
ARBs (e.g., losartan, valsartan)	Block AT1 receptor	Similar to ACEi; NO bradykinin effect $\rightarrow$ no cough	ACEi-intolerant patients; hypertension; HFrEF
MRAs / Aldosterone antagonists (spironolactone, eplerenone)	Block mineralocorticoid receptor	$\uparrow$ Na^+ excretion; $\uparrow$ K^+ retention; anti-fibrotic; anti-androgens (spiro — gynaecomastia, menstrual irregularity)	HFrEF (mortality benefit); Conn's syndrome; oedema; hypertension; PCOS (spiro)
Direct renin inhibitors (aliskiren)	Block renin active site $\rightarrow$ $\downarrow$ Ang I, Ang II	Similar to ACEi; not widely used	Rarely; not used in combination with ACEi/ARB

§10 RENAL ACID-BASE REGULATION

The kidneys are the primary regulators of acid-base balance over hours to days (compared to respiratory buffering which acts in seconds to minutes). The kidney regulates acid-base status by: (1) reabsorbing filtered HCO_3^- (preventing alkali loss), (2) secreting H^+ as titratable acid (phosphate and sulphate buffers), and (3) secreting H^+ as ammonium (NH_4^+). Together, mechanisms 2 and 3 regenerate new HCO_3^- to replace that consumed buffering non-carbonic acids.

Process	Site	Amount/day	Mechanism
HCO_3^- reabsorption	PCT (80–90%), TAL (5%), DCT/CD (5–10%)	4,500 mmol/day reabsorbed	H^+ secreted via NHE3 (PCT) $\rightarrow$ combines with filtered $\text{HCO}_3^- \rightarrow \text{H}_2\text{CO}_3 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ (carbonic anhydrase IV luminal) $\rightarrow$ CO_2 diffuses into cell $\rightarrow$ carbonic anhydrase II $\rightarrow \text{H}_2\text{CO}_3 \rightarrow \text{H}^+ + \text{HCO}_3^- \rightarrow \text{HCO}_3^-$ exits basolateral via NBC1
Titratable acid secretion	Distal tubule and CD	~10–40 mmol/day	H^+ secreted (H-ATPase in type A intercalated cells) $\rightarrow$ titrates $\text{HPO}_4^{2-} \rightarrow \text{H}_2\text{PO}_4^-$ (excreted). Also sulphate, urate, citrate. pH of urine can be lowered to ~4.5 (minimum urine pH).
Ammonium (NH_4^+) secretion	PCT (synthesis); TAL (recycling); IMCD (final excretion)	~30–60 mmol/day (increases 10× in acidosis)	PCT: glutamine $\rightarrow$ glutamate $\rightarrow \alpha\text{-KG} + 2\text{NH}_4^+ + 2\text{HCO}_3^-$ (new HCO_3^- generated); NH_4^+ secreted into lumen via NHE3 (K^+ substitute). TAL: NH_4^+ reabsorbed via NKCC2 $\rightarrow$ medullary accumulation $\rightarrow$ IMCD: NH_4^+ trapped in acid urine

Acid-Base Disorder	pH	Primary Change	Compensation	Renal Response
Metabolic Acidosis	<7.35	↓ HCO ₃ ⁻ (e.g., DKA, lactic acidosis, AKI, diarrhoea)	Hyperventilation (↓ PaCO ₂ — Kussmaul breathing)	↑ H ⁺ secretion; ↑ NH ₄ ⁺ synthesis; ↑ new HCO ₃ ⁻ generation (days)
Metabolic Alkalosis	>7.45	↑ HCO ₃ ⁻ (e.g., vomiting, loop/thiazide diuretics, Conn's)	Hypoventilation (↑ PaCO ₂)	Spill HCO ₃ ⁻ in urine; ↑ renal HCO ₃ ⁻ excretion (if not Cl ⁻ depleted)
Respiratory Acidosis	<7.35	↑ PaCO ₂ (e.g., COPD, hypoventilation, narcosis)	Renal: ↑ HCO ₃ ⁻ reabsorption (2-5 days full compensation)	↑ HCO ₃ ⁻ reabsorption; ↑ H ⁺ secretion
Respiratory Alkalosis	>7.45	↓ PaCO ₂ (e.g., hyperventilation, hypoxia, anxiety, altitude)	Renal: ↓ HCO ₃ ⁻ reabsorption; ↑ HCO ₃ ⁻ excretion	↓ HCO ₃ ⁻ reabsorption; allows HCO ₃ ⁻ loss in urine

§11 RENAL ENDOCRINE FUNCTIONS

Hormone	Stimulus	Synthesis/Action	Clinical Note
Erythropoietin (EPO)	Renal hypoxia (detected by HIF-1α-dependent pathway in peritubular interstitial cells)	Produced by peritubular fibroblasts in renal cortex/outer medulla; acts on erythroid progenitors in bone marrow → ↑ RBC production	CKD → ↓ EPO → normocytic normochromic anaemia of CKD; treat with recombinant EPO (epoetin/darbepoetin); EPO doping in athletes
Renin	↓ Renal perfusion; ↓ NaCl at macula densa; ↑ SNS (β1)	Produced by juxtaglomerular cells (modified smooth muscle of afferent arteriole); cleaves angiotensinogen → Ang I	Renin inhibitors (aliskiren); renin activity measured by plasma renin activity (PRA) or direct renin assay
Calcitriol (1, 25-OH ₂ -D ₃)	PTH; ↓ Ca ²⁺ ; ↓ phosphate; ↓ FGF-23	1α-hydroxylase in PCT converts 25-OH-D ₃ to active calcitriol; FGF-23 (from bone) inhibits 1α-hydroxylase and stimulates 24-hydroxylase (inactivation pathway)	CKD → ↓ 1α-hydroxylase → ↓ calcitriol → hypocalcaemia → ↑ PTH → renal osteodystrophy; treat with alfacalcidol/calcitriol
Prostaglandins (PGE ₂ , PGI ₂)	Ang II; sympathetic stimulation; renal ischaemia	Synthesised locally in afferent arteriole and mesangial cells; vasodilatory; counterbalance vasoconstrictors	NSAIDs block COX → ↓ PGs → constriction uncountered → ↑ risk AKI in susceptible (elderly, HF, CKD, volume-depleted)

§12 CLINICAL CORRELATES

12.1 Acute Kidney Injury (AKI)

Category	Cause	Key Features	Urinary Indices
Pre-renal (60–70%)	Reduced renal perfusion: haemorrhage, dehydration, cardiac failure, sepsis, NSAIDs, ACEi	Oliguria; concentrated urine; BUN/creatinine ratio >20:1; responds to fluid	Low Na ⁺ (<20 mmol/L); High osmolality (>500); FENa <1%; Low urinary Na
Intrinsic Renal (25–35%)	ATN (ischaemia, aminoglycosides, cisplatin, contrast); GN; interstitial nephritis; rhabdomyolysis	Non-oliguric possible; tubular casts in urine; slow recovery (weeks)	High Na ⁺ (>40 mmol/L); Low osmolality (<350); FENa >2%; Granular/muddy brown casts
Post-renal (Obstructive, 5–10%)	BPH; renal calculi; retroperitoneal fibrosis; bladder/cervical cancer	Anuria; may have initial polyuria after relief; bilateral obstruction required for AKI (unless single kidney)	Variable; dilute once established

12.2 Nephritic vs Nephrotic Syndrome

Feature	Nephritic Syndrome	Nephrotic Syndrome
Pathological mechanism	Inflammatory glomerular damage → ↑ permeability to cells and some protein	Non-inflammatory damage to filtration barrier → ↑ permeability to large proteins
Proteinuria	Mild-moderate (<3.5 g/day)	Heavy (>3.5 g/day; may be 20–30 g/day)
Haematuria	Hallmark — dysmorphic RBCs, RBC casts	Absent or mild
Oedema	Mild	Massive (periorbital, ascites, pulmonary, peripheral)
Hypertension	Common (Na ⁺ + water retention)	Variable
Hypoalbuminaemia	Absent or mild	Hallmark (albumin lost in urine)
Hyperlipidaemia	Absent	Yes (↑ hepatic lipoprotein synthesis)
Examples	IgA nephropathy (Berger's); post-streptococcal GN; ANCA vasculitis; SLE (class III/IV); Goodpasture's	Minimal Change Disease (children); FSGS (adults); Membranous GN; Diabetic nephropathy; Amyloidosis

USEFUL EQUATIONS IN RENAL PHYSIOLOGY

GFR (Clearance method): $GFR = (U \times V) / P$, where U = urinary concentration, V = urine flow rate, P = plasma concentration (for freely filtered, unreabsorbed, unmetabolised substance like inulin or creatinine).

Fractional Excretion of Na^+ (FENa): $FENa = (UNa \times PCr) / (PNa \times UCr) \times 100\%$. $<1\%$ = pre-renal; $>2\%$ = intrinsic AKI.

Filtration Fraction (FF): $FF = GFR / RPF = \sim 125 / 625 = 20\%$. High FF $\rightarrow$ more protein concentrated in efferent $\rightarrow$ $\uparrow$ peritubular oncotic pressure $\rightarrow$ $\uparrow$ PCT reabsorption.

Anion Gap: $AG = [Na^+] - ([Cl^-] + [HCO_3^-])$ = normally 8–12 mEq/L. Elevated AG in high anion gap metabolic acidosis (MUDPILES: Methanol, Uraemia, DKA, Propylene glycol, Isoniazid/Infection (sepsis), Lactic acidosis, Ethylene glycol, Salicylates).